

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Examination – Summer 2024

Course: B. Tech. Branch: CIVIL Semester: V
Subject Code & Name: Advance Environmental Engineering BTCVPE506A
Max Marks: 60 Date: 12/07/2024 Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		
A) Design a septic tank for 75 users. Assume necessary data.	CO I/BT V	06
B) Rewrite the equation that is used for modelling of self-purification of stream process. Explain Each and Every term used in it.	CO I/BT VI	06
C) Classify Low-cost treatment methods. Explain Bacterial Algal Symbiosis in detail with the help of neat sketch.	CO I/BT IV	06
Q.2 Solve Any Two of the following.		
A) Explain effect of industrial waste discharge on STPs.	CO II/BT IV	06
B) Explain processes of neutralization of Alkaline industrial wastewater. Discuss any two methods.	CO-II/BT IV	06
C) Explain General Characteristics of Industrial Wastewater.	CO II/ BT IV	06
Q. 3 Solve Any Two of the following.		
A) Consider following data: A industry wants to recycle the industrial wastewater containing heavy metals like Hg and Pb. Industry has enough funding to choose best suited technology as a tertiary treatment. Industry authorities wish to have recycled water quality to be amongst best. Propose a suitable heavy metal removal technology for secondary treated waste of above industry.	CO-III/ BT VI	06
B) Compare Nitrification and Denitrification of wastewater	CO-III/ BT V	06
C) Explain how Ion Exchange can be used for tertiary treatment of Industrial Wastewater.	CO-III/BT IV	06
Q.4 Solve Any Two of the following.		
A) Categorize vector Transmitted diseases. Explain any one in detail.	CO-III/ BT V	06
B) Explain Life cycle of house fly. Also examine fly control	CO-III/BT IV	06

w.r.t. different stages of life cycle

C) **Explain** chain of disease transmission CO -III/BT IV **06**

Q. 5 Solve Any Two of the following.

A) **Examine** suitability of septic tanks in environmental sanitation with help of neat sketch. CO-IV/BT VI **06**

B) **Explain** water quality criteria prescribed by Central Pollution Control Board (CPCB) CO-IV/BT IV **06**

C) **Evaluate** suitability of Eco-San toilet. Also draw a neat sketch CO-IV/BT VI **06**

***** End *****